

Topic 9: AI Capabilities and Limitations

9.1 What Generative AI Does Well

- Language tasks: summarising, translating, classifying, drafting, rewriting, explaining -- often at or above human level
- Code generation: writing boilerplate, explaining code, finding bugs, suggesting tests, converting between languages
- Pattern recognition across modalities: classifying images, transcribing speech, describing visual content
- Knowledge retrieval: answering questions about well-documented topics present in training data
- Creativity support: brainstorming, ideation, generating variations, producing first drafts
- Speed: processes and generates thousands of words in seconds

9.2 Hallucinations

- Hallucination: the model generates confident, fluent, but factually wrong information
- Root cause: LLMs predict the most statistically plausible next token, not verified truth -- plausible != accurate
- Common hallucinations: invented citations, wrong dates, fictional people presented as real, incorrect statistics
- Hallucination rates vary widely by model, topic, and prompt -- less common on well-documented topics, more on obscure ones
- Mitigation strategies: RAG (ground the model in verified documents), ask the model to cite sources, always verify critical facts independently
- Rule: never trust AI output on specific factual claims without independent verification

9.3 Bias, Fairness, and Safety

- AI models learn the biases present in their training data -- human societal biases become model biases
- Gender bias: models may default to male pronouns for roles historically dominated by men
- Racial bias: some image generation models produce stereotyped representations of ethnic groups
- Cultural bias: models trained predominantly on English-language content perform worse for other languages and cultures
- Safety fine-tuning reduces but does not eliminate harmful outputs -- models can still be misused
- Ongoing research areas: bias auditing, red-teaming, constitutional AI, and diverse training data curation

9.4 Fundamental Limitations

- Knowledge cutoff: models have a training data cutoff date and do not know about events after it (without tool use or retrieval)
 - No persistent memory: each conversation starts fresh unless memory is explicitly implemented at the application level
 - Arithmetic and precise reasoning: LLMs can make arithmetic errors; use code execution for reliable calculations
 - Long-context degradation: performance can degrade on tasks that require using information from very early in a long context
 - Consistency: the same prompt can produce different outputs across runs due to temperature (sampling randomness)
 - No real-world agency: a base LLM cannot browse the internet, run code, or take actions without external tools
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